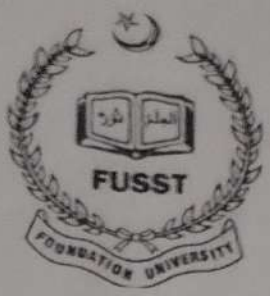
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Foundation University School of Science & Technology

TOPIC

Pumping Lemma and DFA

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i) Assume L is regular.
if L is regular, Pumping Lemma applies.
There exists a pumping length p.

ii) Choose a string in L.
Let

$$s = a^{2p} b^{2p}$$

$s \in L$ and $|s| \geq p$.

iii) Split the string:-
According to Pumping Lemma:

$$s = xyz$$

where $|xy| \leq p$
$|y| \geq 1$

first p symbols are only a's:

$$y = a^k \quad (k \geq 1)$$

iv) Pumping Down ($i=0$):

$$xy^0z = a^{2p-k} b^{2p}$$

v) Contradiction:

No. of a's $\neq$ No. of b's.

$\therefore xy^0z \notin L$

Proved, $L = \{a^{2n} b^{2n}\}$ is NOT regular.

THE END.

1) Write regular expression for the language of all binary strings that contain at least two consecutive.

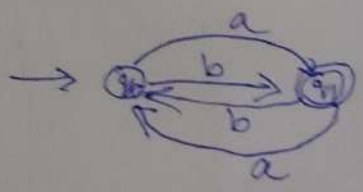
Regular Expression:-

$$= (0+1)^* 11 (0+1)^*$$

Accepted = {11, 011, 0110, 1011, ---}

2) Construct a DFA that accepts all strings over {a, b} where the no. of a's is even.

DFA Construction:-



State	a	b
q_0	q_1	q_0
q_1	q_0	q_1

q_0 = accepting state
q_0 = start state.

3) Use the Pumping Lemma to prove that the language $L = \{a^{2n} b^{2n}\}$ is not regular.

Pumping Lemma Proof:-

$$L = \{a^{2n} b^{2n} \mid n \geq 0\}$$